Intro to Numerical Methods

3rd semester

What are Numerical Methods?

- Not all mathematical problems are same. Some are simple and some are complex.
- But a digital computer are simpler machine in low-level.
- Complex problems require specialized optimizations, approximations, and situational heuristics, etc.
- So, any problems which are solved by building approaches by “converting” those problems to be “solvable” using basic arithmetic operations can be said to be a numerical method/computing.
- “**Numerical method/computing is an approach for solving complex mathematical problems using only simple arithmetic operations. The approach involves formulation of mathematical models of physical situations that can be solved with arithmetic operations.**”

When and Where NM?

- When closed-form solutions are difficult or impossible (e.g. gradient descent)
- When problems have solutions but are complex.
- Where: real-world engineering, physics, finance, graphics, AI, etc.

What troubles NM?

- Large number of arithmetic calculations.
- So require fast and efficient computing devices.

Some example of NM usage:

1. Finding roots of equations
2. Solving systems of linear algebraic equations
3. Interpolation and regression analysis
4. Numerical Integration
5. Numerical Differentiation
6. Solution of differential equations
7. Boundary value Problems
8. Solution of matrix Problems

Process of Numerical Computing

- Numerical computing involves formulation of mathematical models of physical problems that can be solved by using basic mathematical operations.
- The problem of numerical computing can be roughly divided into the following four phases:
  - Formulation of mathematical model.
  - Construction of approximate numerical methods.
  - Implementation of method to obtain a solution.
  - Validation of the solution.

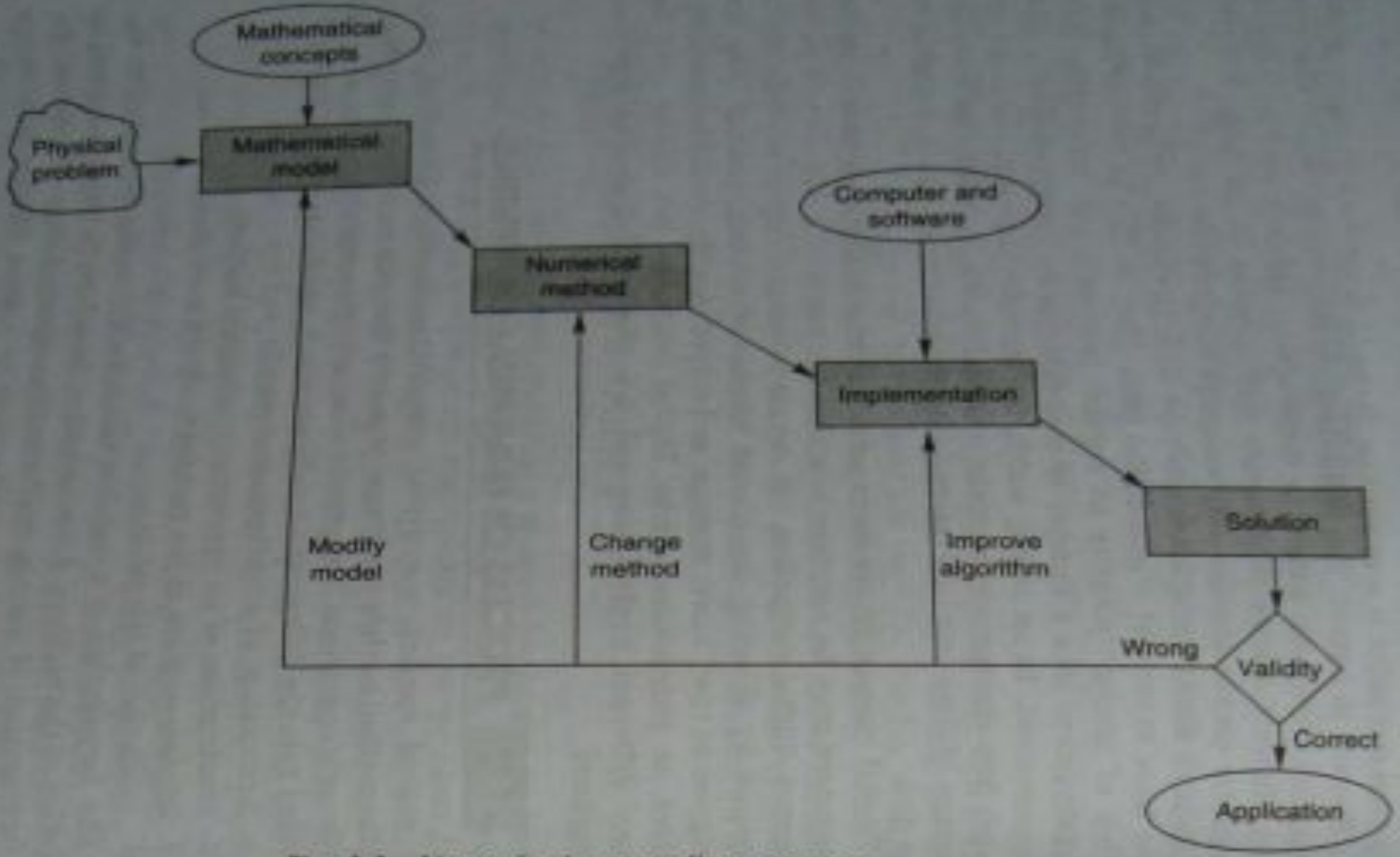

Fig. 1.1 Numerical computing process

Characteristics of Numerical Computing:

1. Accuracy - The results we obtain must be sufficiently accurate to serve the purpose for which mathematical model was build.
2. Efficiency - A method that requires less computing time and less programming effort and yet achieves the desired accuracy is always preferred.
3. Rate of convergence - Many numerical methods are based on the idea of an iterable process. This process involves generation of a sequence of approximations with the hope that the process will than others. Some methods may not converge at all. It is therefore important to test for convergence before a method is used.

4. Numerical Stability - In some cases errors tend to grow exponentially with disastrous computational result. A computing process that shows such exponential growth is said to be numerically unstable. We must choose methods that are not only fast but also stable.

Some history on NM

- Numerical Methods predates Digital Computers. E.g. Linear Interpolation.
- Some pre-computing algorithms are Newton's method, Gaussian elimination and Euler's method.
- The first modern methods often is credited to John von Neumann and Goldstine paper in 1947.

Error

- The difference between the exact solution of a problem and the approximate solution obtained through a numerical method.
- In numerical methods, errors will arise during the calculations due to the finite number of operations a computer can do.
- We deal with them by identifying the errors, quantifying it and lastly minimizing the error as per our needs.

Types of Error

1. True Error:

   Difference between the true value (exact value) and the approximate value.

$$E_t = \text{True value} - \text{Approximate value}$$

2. Relative Error:

   Defined as the ratio between the true error and the true value.

$$\epsilon_t = \frac{\text{True Error}}{\text{True Value}}$$

Types of Error:

3. Approximate Error - The difference between the present approximation and previous approximation

$$E_a = \text{Present Approximation} - \text{Previous Approximation}$$

4. Relative Approximation Error - The ratio between the approximation error and the present approximation

$$\epsilon_a = \frac{\text{Approximate Error}}{\text{Present Approximation}}$$

Sources of Error

There are multiple ways for errors to exist in the numerical model.

- Errors in the model itself, errors from mistakes in programs themselves or in measurement of physical quantities.
- Broadly, errors are of following kinds:

1. Round off error:
    - A number can be represented upto a certain digits.
    - Round off errors occur when a fixed number of digits are used to represent numbers.
    - E.g. ⅓ is 0.333333 on PC. then the round off error would be: ⅓ -0.333333 = 0.00000033.
    - There are irrational numbers like π and √2.

Sources of Error

2. Truncation Error:

- Truncation error is defined as the error caused by truncating mathematical procedure.
- For e.g., the Maclaurin series for e^x is given as: $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \ldots$

Although, this is a infinite series with infinite number of terms, to calculate e^x, only a finite number of terms

$$e^x - \left(1 + x + \frac{x^2}{2} + \ldots \right)$$

one uses three terms to calculate e^x, then,

$$\frac{x^3}{3} + \frac{x^4}{4} + \ldots$$

So Truncation error will be:

Sources of Error:

3. Relative/Absolute Error: If x denotes the true value of a numerical quantity and x_0 denotes the approximate value then

$$\text{Absolute Error} = | x_t - x_0|$$

$$\text{Relative Error} = (\text{absolute error} / |\text{true error}|)$$

4. Percentage Error - (RE x 100)%

Question

1. We are given an approximate value of π is 3.1428571 and true value is 3.1415926. Calculate true, absolute, relative and percentage errors?

Propagation of Errors:

- If a calculation is made with numbers that are not exact, then the calculation itself will have error.
- Propagation of error is the effect of variables errors on the uncertainty of a function based on them.
- When the variables are the values of experimental limitations which propagate to the combination of variables in the function.